29 final Question

Point out the evaluation criteria that is used in Evaluation techniques. Explain Heuristic evaluation techniques through expert analysis

✅Evaluation Criteria Used in Evaluation Techniques

When evaluating a system or interface, certain **criteria** are commonly used to assess **usability, performance, and user satisfaction**. These include:

📋Key Evaluation Criteria:

1. **Learnability**
    - How easily can new users learn to use the system?
2. **Efficiency**
    - How quickly can users complete tasks once they've learned the system?
3. **Memorability**
    - Can users remember how to use the system after a break?
4. **Errors**
    - How many errors do users make?
    - How severe are these errors?
    - Can users recover easily?
5. **Satisfaction**
    - How pleasant is the system to use?
    - Do users feel comfortable and confident?
6. **Effectiveness**
    - Does the system help users achieve their goals accurately?
7. **Flexibility**
    - Can the system be used in a variety of ways by different users?

8. **Feedback**
    - Does the system provide timely and helpful feedback?
9. **Consistency**
    - Is the system's behavior predictable and uniform across all parts?

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Heuristic Evaluation Technique (Expert Analysis)

✅Definition:

Heuristic evaluation is a **usability inspection method** where one or more **experts review a user interface** and evaluate it against a set of **established usability principles** (called **heuristics**), usually those proposed by **Jakob Nielsen**.

🔍

Process of Heuristic Evaluation:

1. **Select Evaluators**
    - Typically 3 to 5 usability experts perform independent evaluations.
2. **Define the Heuristics**
    - Use a predefined set, such as Nielsen's 10 heuristics (e.g., consistency, visibility, error prevention).
3. **Conduct the Evaluation**
    - Experts go through the interface multiple times.
    - Identify violations of heuristics.
    - Note usability problems, context, and severity.
4. **Record Problems and Severity Ratings**
    - Each issue is assigned a **severity score** (e.g., 0 = not a problem, 4 = usability catastrophe).
5. **Aggregate and Analyze Findings**
    - Combine expert feedback.

- Prioritize fixes based on severity and impact.

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Why Use Heuristic Evaluation?

- Quick and cost-effective
- Doesn't require real users
- Good for **early-stage prototypes**
- Identifies issues before large-scale testing

📌

Limitations:

- Depends on evaluator skill
- May miss user-specific issues
- Subjective to expert judgment

Let me know if you want a checklist format for the heuristics or a printable summary! 😊

Differentiate between virtual reality and augmented reality. write some application of augmented reality

✅Difference Between Virtual Reality (VR) and Augmented Reality (AR)

Feature	Virtual Reality (VR)	Augmented Reality (AR)
Definition	A fully immersive digital environment	Overlay of digital content on the real-world environment
Environment	Entirely virtual; user is cut off from the real world	Real world enhanced with digital objects
Devices Used	VR headset, motion controllers, gloves	Smartphones, tablets, AR glasses (e.g., HoloLens)

Interaction Level	High immersion and full interaction in the virtual space	Limited interaction, mostly enhancements of real world
Example	VR gaming (e.g., Beat Saber), VR training simulations	Pokémon GO, IKEA Place app, Snapchat filters
Use Case Focus	Simulation, training, entertainment	Navigation, real-time information, design, education

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✅Applications of Augmented Reality (AR)

1. **Education**
    - Interactive learning using 3D models (e.g., anatomy, astronomy).
    - AR textbooks and flashcards for immersive learning.
2. **Healthcare**
    - Visual overlays for surgeries or diagnostics.
    - Training for medical students using simulated AR environments.
3. **Retail & E-commerce**
    - Try-before-you-buy features (e.g., virtual fitting rooms, AR furniture placement by IKEA or Amazon).
4. **Gaming & Entertainment**
    - Games like **Pokémon GO**, AR escape rooms, and interactive museum tours.
5. **Navigation & Travel**
    - Real-time directions and landmark identification using AR (e.g., Google Maps AR walking mode).
6. **Manufacturing & Maintenance**
    - Technicians can view step-by-step instructions overlaid on machinery.
    - Remote assistance using AR headsets.

Let me know if you'd like this information in table format or a visual summary! 😊